



Smart Grid is less about smart technologies. It is about smart business decisions to invest into the right technologies, aimed at providing affordable, reliable electricity to the consumer —Sandra Diethelm, CEO, Exuta (India) Pvt Ltd

Definitions

- AMI: Advanced metering infrastructure
- CTU: Central transmission utility
- DMS: Distribution management systems
- GIS: Graphical information systems
- ISGF: India Smart Grid Forum (by MoP)
- MDMS: Meter data management systems
- PV2G: Photovoltaic to grid
- R-APDRP: Restructured-Accelerated Power Development and Reforms Programme
- V2G: Vehicle to grid

understanding of smart grid solutions in the energy field.

The Power Grid Corporation of India Limited (POWERGRID) has a 1200kV ultra high voltage (UHV) AC national test station, which is the world's highest voltage level, located at Bina in Madhya Pradesh. The station provides a ground to test new equipment and technologies at a voltage level never seen before. Manufacturers worldwide have flocked to Bina to jointly explore whether 1200kV is a feasible new technology.

Incoming investments. BORG Energy India will invest US\$ 45 million in Indian market for rural electrification and engineering, procurement and construction (EPC) projects within six months after January 2014. BORG is also planning to set up an R&D facility in 2016 towards nanotechnology in solar panels, said Joseph Fournier, director, BORG Inc at a press meet on 13th January.

CG has invested ₹ 80 million in a manufacturing facility for smart grid devices, which has been set up in Bangalore and can employ more than 100 people. The smart grid facility will manufacture substation automation products, distribution automation devices, protection and control systems, advanced metering infrastructure (AMI) and telecommunication

solutions. It will also offer global engineering services such as systems integration, installation and commissioning. The facility is fully equipped with modern equipment to ensure an annual production capacity of 10,000 units of power line carrier communication terminals (PLCCs) and intelligent electronic devices (IEDs).

Trends to look out for

Whilst the integration of renewable energy systems into existing grids will help cater to higher demand and improving efficiency, it will also need to consider the challenges posed by new consumption models like EVs and home automation. Additionally, increased use of sensors into different aspects of the grid, such as transformers, allows analysts to gain more data from the grid that they can then use for forecasting. "Furthermore, with the internet of things (IoT) enablement, smart meters and homes will come into play, which will further help optimise energy consumption," adds Shailesh Thakurdesai, business development manager—microcontrollers, Texas Instruments, India.

"Based on the feedback that we received, this market also requires an extended temperature operating range of -40°C to 70°C and shares a need for secure Ethernet, redundant power supply, additional serial ports and running these new protocols such as DNP3 or IEC 61850," explains Jamie Smith, director, Embedded Systems in a previous interview with EFY.

Microgrids are another popular trend. Enabling local power generation, microgrids comprise various small power-generating sources that make them highly flexible and efficient. Talking about this, "We have integrat-

ed existing hardware components in the market with a powerful embedded system. The setup forms an electricity grid that is small enough to be called a microgrid. This system is powered by renewable energy sources such as solar photovoltaics (PVs), wind and micro-hydel. It can be remotely monitored from anywhere in the world," explains Ganesh Shankar, managing director, FluxGen Engineering Technologies.

One smart grid application example is with LocalGrid Technologies, which focusses on a microgrid application where they are controlling the energy coming on and off the grid. "This application is able to provide security features because it is based on Linux, and the additional performance made this a more viable solution for the smart grid compared to that achievable with the NI cRIO-9074 system," adds Smith.

Government policies promoting smart grid implementation

B. N. Sharma, IAS, joint secretary, Ministry of Power, states in the roadmap that APDRP-I upgraded basic electricity networks in 56 urban circles, whereas RGGVY electrified 56 million households. The ongoing R-APDRP covers 1401 towns across state-owned electricity distribution companies. Under this programme, the distribution companies are building their IT networks and a set of basic IT and automation systems for the first time.

The Kerala State Electricity Regulatory Commission (KSERC) has put in place a system to enable large power consumers to access energy from sources of their own choice, using the linked network of transmission and distribution licencees.

Power Grid Corporation of India jointly with Electricity Department, government of Puducherry, is developing a smart grid pilot project at Puducherry through open collaboration. The project is being implemented to achieve a common information-sharing platform, scalable and replicable at other places, and shall also help in